

Analysis of IoT Botnet Structure and Behavior: A Study Based on the Hitori (SlumX) Variant

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SlumX: 12 Architectures • 7 Attack Vectors • 1 Sophisticated Threat

The Evolution from Simple Mirai Variants to Advanced Multi-Platform IoT Weaponry

1. Abstract

- Technical analysis of **SlumX botnet** (Hitori variant) via static code analysis [1]
- Sophisticated **C2 mechanism** with diverse DDoS methods (UDP, TCP, HTTP)
- Multi-architecture support**: MIPS, ARM, x86, and 9 more platforms [2]
- Advanced **evasion techniques** unique to IoT devices
- Contribution**: Documentation of modern IoT botnet behaviors and mitigation strategies [5].

2. Research Objectives

General Objective

Conduct an in-depth technical study of the Hitori (SlumX) botnet variant to better understand the threat posed by modern IoT botnets [1].

Specific Objectives

- Investigate the underlying code architecture to determine major functions and operation.
- Identify and assess DDoS attack techniques and payload delivery methods [3].
- Evaluate multi-architecture and cross-platform support mechanisms.
- Recommend mitigation methods for discovered vulnerabilities [4].

3. Methodology

Phase	Description
Data Collection	GitHub repo <code>mat1520/slumpx</code>
Static Analysis	<code>bot.c</code> , <code>server.c</code> , <code>cc.py</code>
Control Flow	Execution paths mapping
Threat Modeling	Attack surface assessment

Note: Static analysis only - no runtime execution [2].

4. Threat Model

12+

Supported Architectures

7+

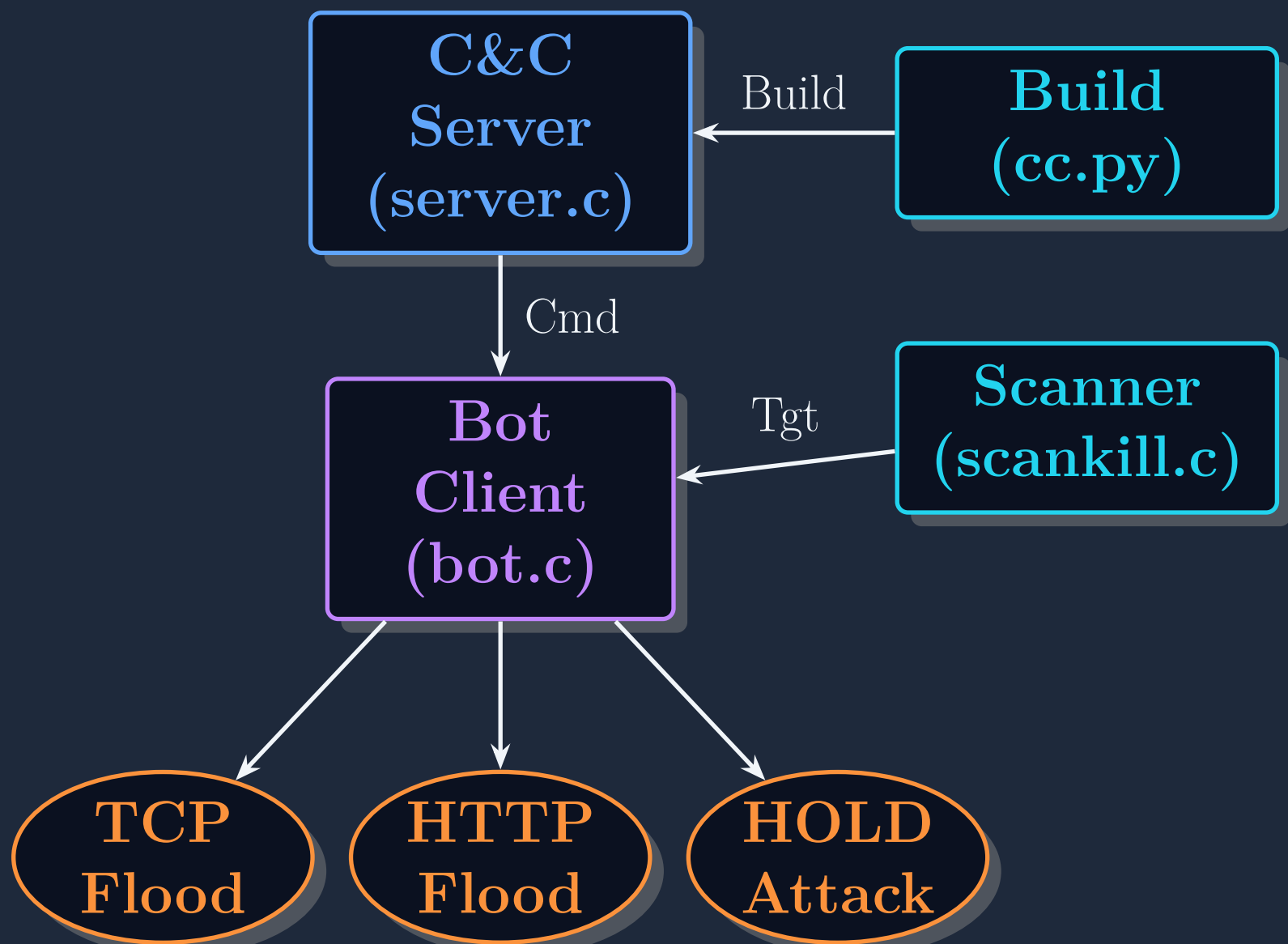
Attack Vectors

1Tbps+

DDoS Potential

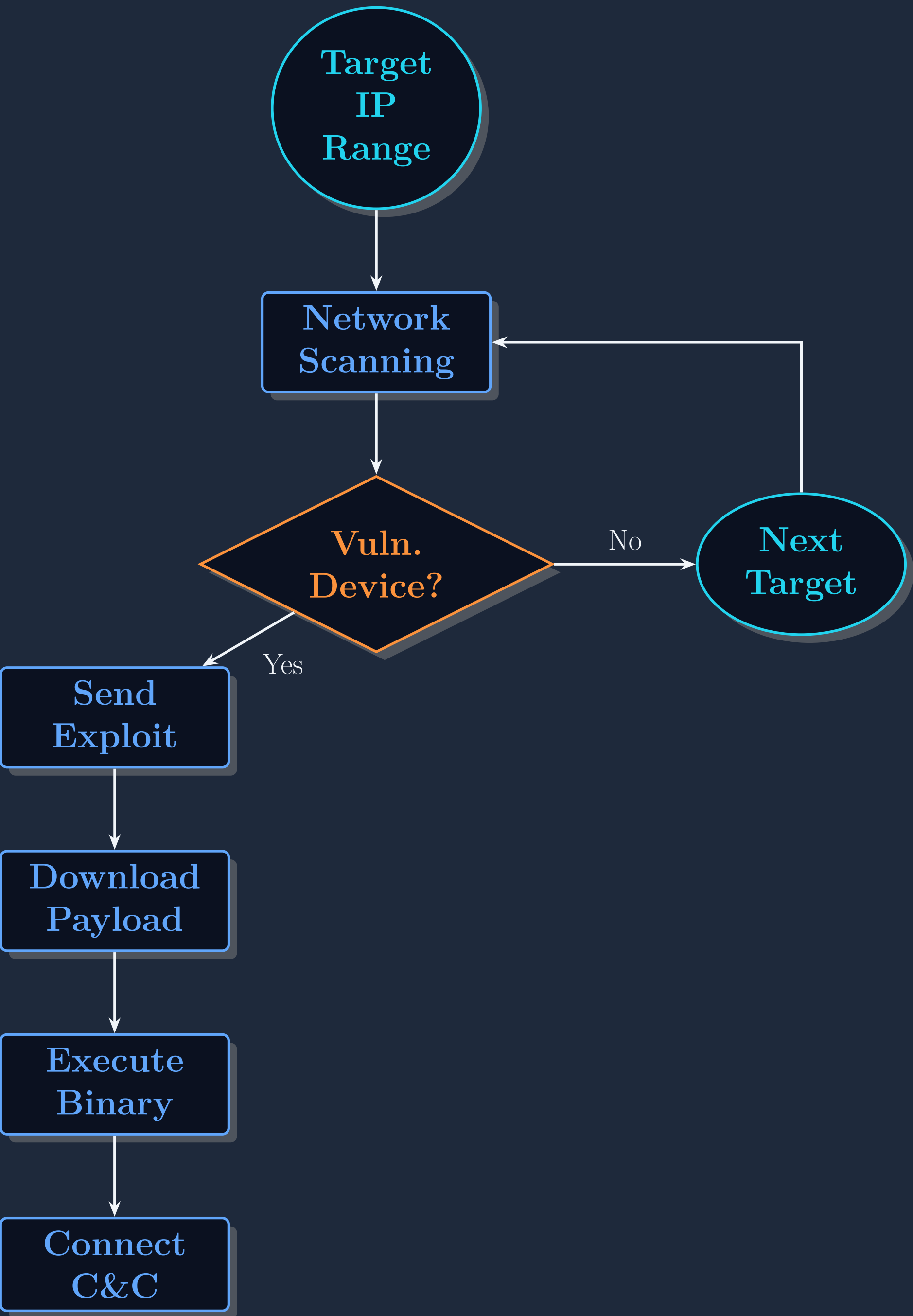
Vulnerabilities: Buffer overflows, no thread sync, hardcoded credentials.

5. System Architecture



The architecture follows a centralized C2 model [1]. The `server.c` component manages the botnet via TCP port 1994, while `bot.c` resides on infected devices. The `cc.py` build system is crucial for cross-compiling payloads for various architectures.

6. Infection Process Flow



The infection cycle begins with network scanning (`scankill.c`) to identify vulnerable devices. Upon successful exploitation (e.g., via Telnet brute-force or specific CVEs), the payload is downloaded and executed, establishing a persistent connection to the C2 server [2].

7. Security Risk Assessment

Hardcoded C&C (Single Point of Failure)	Plain Text Communication (Traffic Analysis)
Fixed Port Usage (1994) (Filtering)	Predictable Binary Names (Signatures)

High Risk: Hardcoded IP addresses allow defenders to sinkhole the entire botnet. **Medium Risk:** Lack of encryption enables traffic inspection and signature generation.

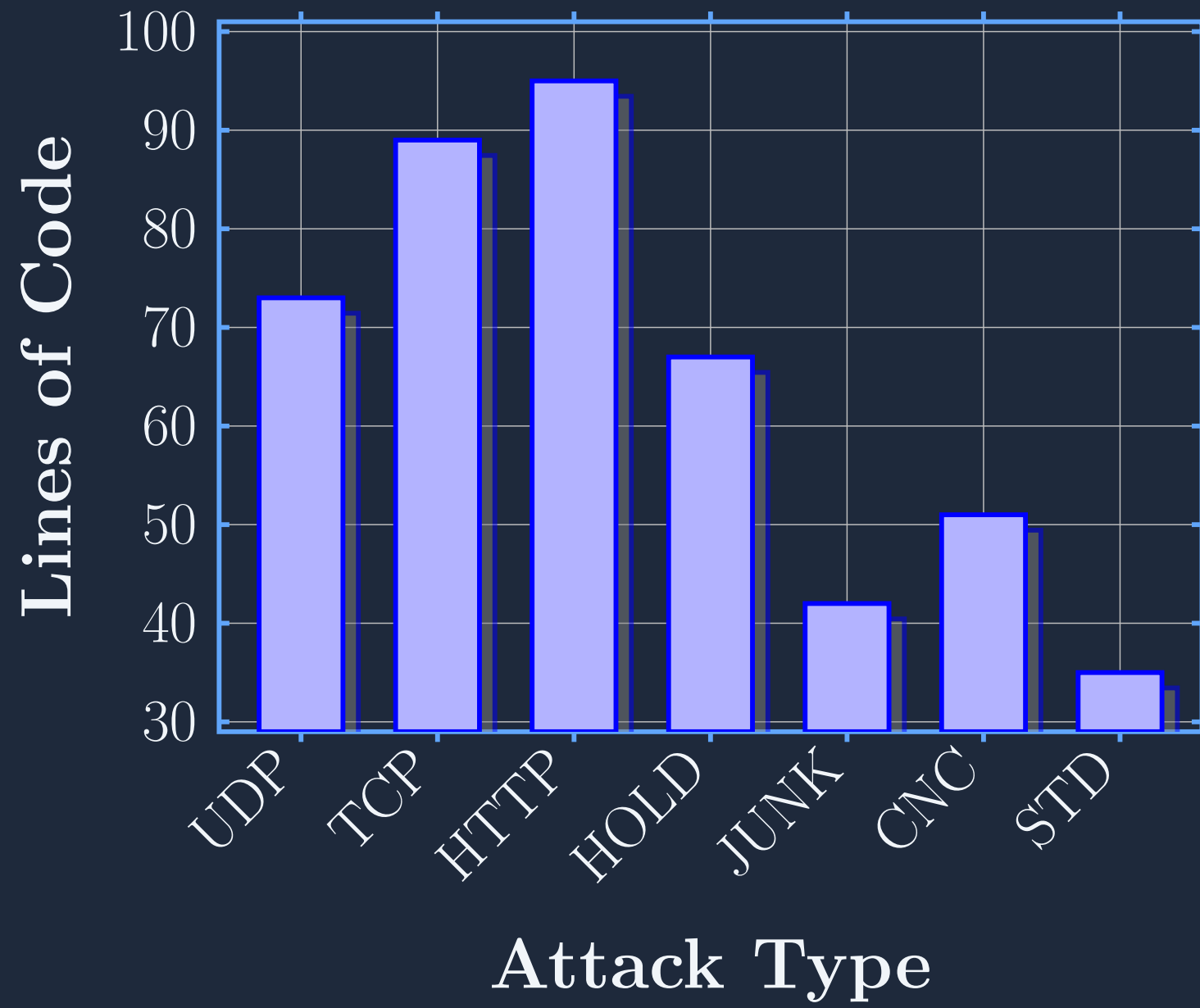
8. Multi-Architecture Support

Architecture	Binary	Target Devices
MIPS	8mips8	Router firmware, embedded systems
MIPSEL	8mps18	Little-endian MIPS devices
SH4	8sh48	SuperH architecture devices
i586	8i5868	Legacy x86 systems
ARMv6l	8arm68	ARM-based IoT devices
i686	8i6868	Modern x86 systems
PowerPC	8ppc8	PowerPC-based systems
M68k	8m68k8	Motorola 68000 systems
SPARC	8spc8	SPARC architecture systems
ARMv4l	8arm48	Older ARM devices
ARMv5l	8arm58	ARMv5 systems
ARMv7l	8arm78	Modern ARM devices

12 architectures supported - Massive IoT attack surface

Unlike early IoT botnets, SlumX supports 12 distinct architectures, significantly increasing the potential infection pool [3][5].

9. Attack Vector Complexity



HTTP-based attacks show highest complexity (95 LOC).

The botnet supports multiple DDoS vectors. Analysis shows that HTTP floods are the most complex to implement (95 LOC), requiring valid header construction to bypass simple filters. In contrast, UDP and TCP floods rely on raw socket manipulation for high-volume traffic generation [3].

10. Key Findings & Conclusions

- Multi-architecture:** 12 architectures (MIPS, ARM, x86, embedded) dramatically expanding attack surface.
- Advanced evasion:** Process masquerading, anti-analysis features, IoT-specific hiding techniques.
- Targeted exploitation:** Vendor-specific exploits showing evolution to targeted attacks.
- Future Trends:** Mirai-driven botnets are moving toward more sophisticated infrastructures with increased persistence [4].

Future Research:

- Dynamic analysis frameworks
- Specialized detection technologies [4]

References

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